

wavelength stability, turn-on delay, power dissipation, etc.

VCSELs. VCSELs have made progress recently, especially with the advent of an oxide confined structure that promises to lower threshold currents. Compared to quantum-well modulators, VCSELs have seen less use yet in systems with large arrays operating with the silicon circuit.

Photodetector and receiver circuit. Detector is a very important part for the receiver's performance as it converts optical energy into electrical energy. For an on-chip optical interconnect system, the detector should have a low input capacitance if the receiver circuit is to be kept small with not too large power dissipation. Large capacitances mean more sensitive amplifiers have to be used, which introduce a noise that degrades the performance of the receiver circuit while also increasing the power dissipation.

A better approach for optical inter-

connect receivers is to make the physical capacitance of the photodetector and its connection to the receiver circuit as small as possible. Small capacitance leads to larger voltage swings for a given optical energy, which leads to better noise immunity and improved gain stages. Smaller capacitance also allows use of small, low-power-dissipation transistors in the input stage. To calculate the total capacitance, you need to estimate the gate capacitance of the transistor with which the detector would be integrated. For 90nm CMOS technology node the gate capacitance of an NMOS transistor per unit transistor width is estimated to be $2 \text{ fF}/\mu\text{m}$, and for the 32nm node the corresponding number is $1.2 \text{ fF}/\mu\text{m}$.

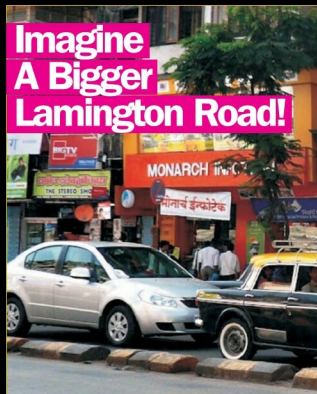
To sum up

We have seen various advantages of the optical interconnect system over the electrical interconnect. Optical interconnects solve many of the problems arising in electrical interconnects

due to chip-to-chip delay, crosstalk noise, bandwidth, interconnect wire and system synchronisation. Optics may reduce the power dissipation in clock distribution; global on-chip and off-chip interconnects improve the timing. We have found that optical interconnects have a performance that is competitive with or better than electrical interconnects' and can scale to future interconnect needs.

In conclusion, optics is very promising for dense interconnects to silicon chips. Overall, electrical and optical interconnects, and the many existing and emerging technologies in optoelectronics and optics integrated with silicon CMOS, are likely to play a substantial role in solving major problems in scaling interconnects for CMOS chips in the coming decades. ●

Kapil Khare is doing M.Tech in Microelectronics and VLSI Design from NIT, Durgapur, West Bengal. Dr Rajib Kar is an assistant professor in the department of Electronics and Communication Engineering at NIT, Durgapur, West Bengal



**Check Out
India's leading
market-places
for trade &
commerce in
electronics**

eleB2B.com & **ElectronicsForYou**
YELLOW PAGES

